Hypothesis Testing
Differences Among Groups

I. The Idea

II. Analysis of Variance (ANOVA or F Test)

A. Two Estimates of Variance
B. The F Statistic

III. Examples

A. Cell Phone Use and Sperm Motility

BME 423 Discussion - Week 4

Outline

I. Probability

- Examples (by-hand)
- Examples with R probability distribution functions: pnorm(), qnorm()

II. ANOVA

- How does ANOVA work?
- Recap : ANOVA
- Recap : The F Statistic
- Example : Is inappropriate drug prescribing associated with a longer hospital stay?
- In-class exercise : Are mood changes associated with the assigned antidepressant drug groups?
- Guided Tour : aov()

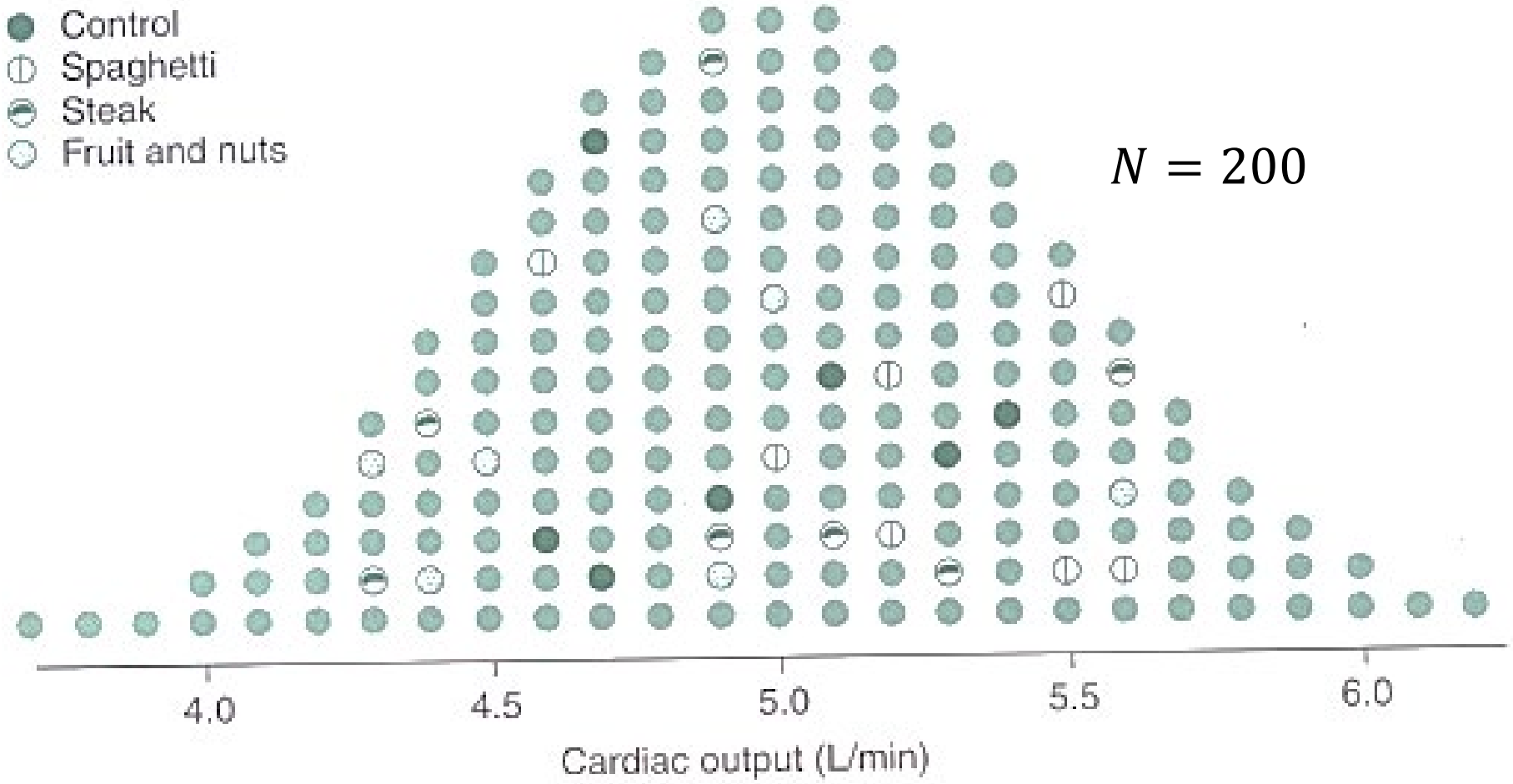
I. The Idea

Hypothetical Experiment on the Effects of Diet on Cardiac Output

- Population of 200 healthy adults
- Randomly select 4 groups of 7 people each
- Each group assigned different diet:
  - Group 1: Normal diet (control)
  - Group 2: Pasta only
  - Group 3: Steak only
  - Group 4: Fruit & nuts
- After 1 month, measure Cardiac Output (CO) of each

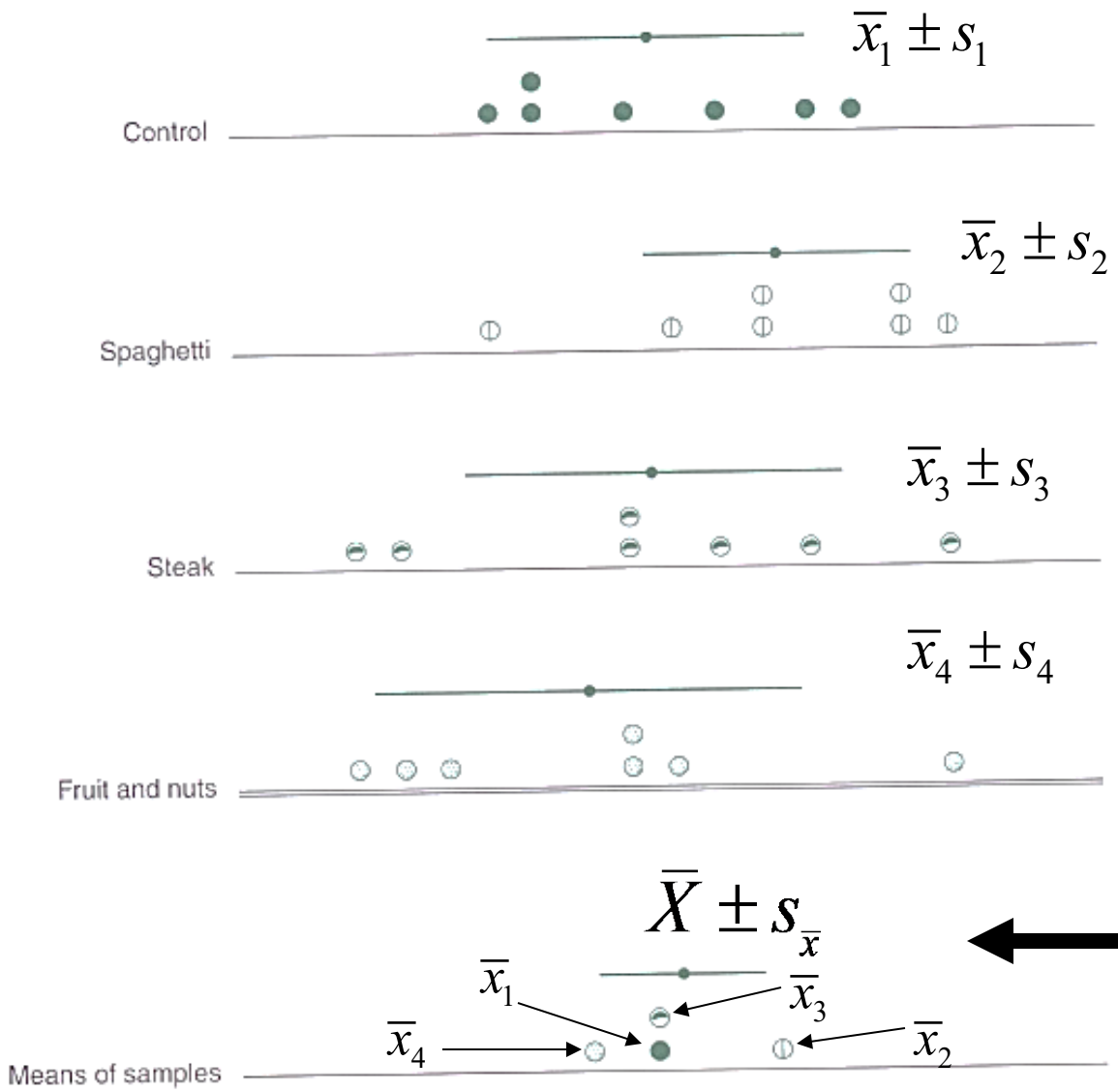
Hypothesis: All diets (treatments) have the same effect on CO (i.e., diet has <u>no effect</u> on CO)

$$X \sim N(\mu, \sigma^2)$$



$N = 200$

If we observe the entire population – Diet has No Effect on CO



Questions:

Are these differences in CO due to diet or are these due to random variation in CO in the population?

Is this variability in $\bar{x}_j$ consistent with the variability observed within the individual sample x_i values?

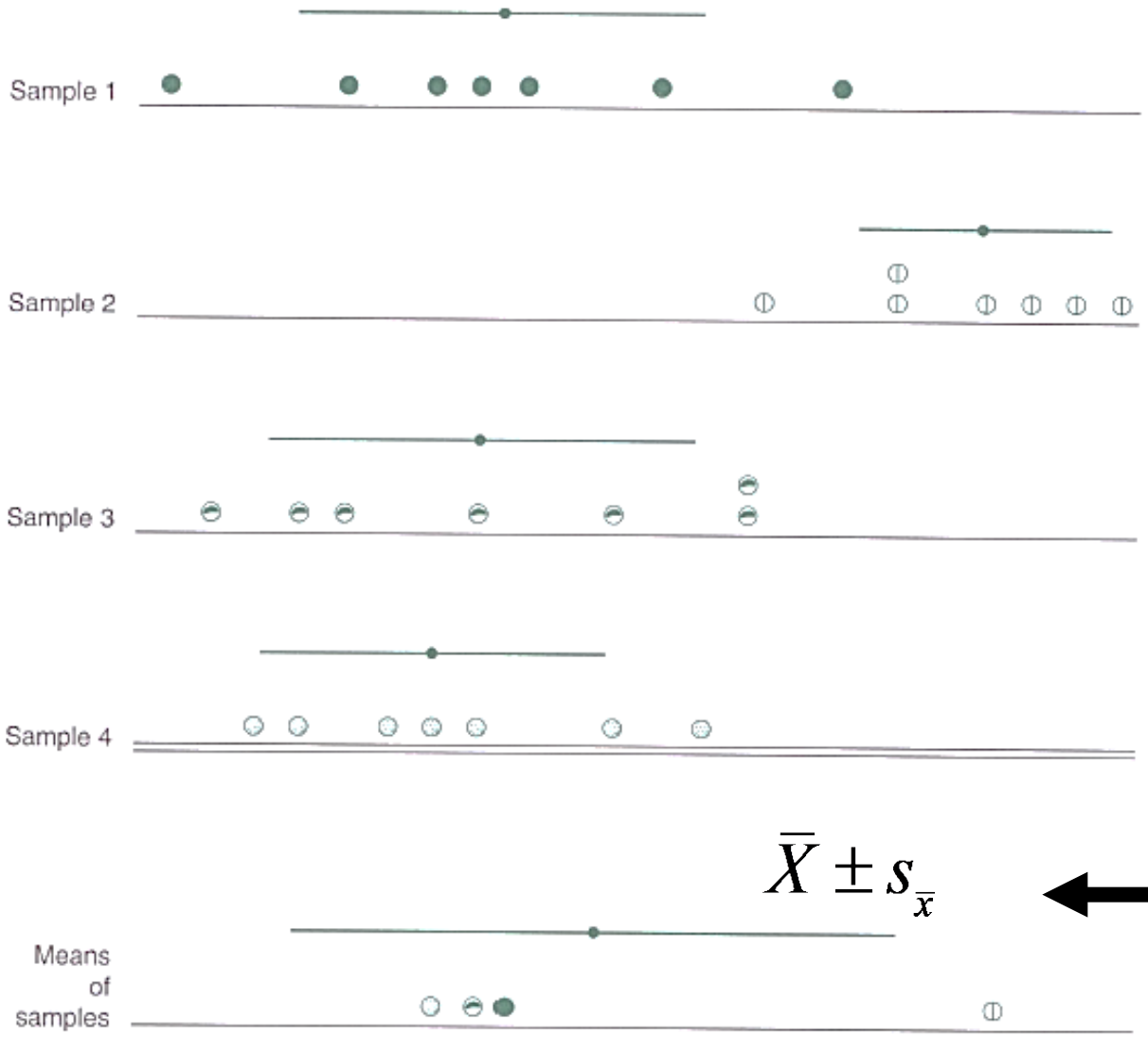


FIGURE 3-3. The four samples shown are identical to those in Figure 3-2 except that the variability in the mean values has been increased substantially. The samples now appear to differ from each other because the variability between the sample means is larger than one would expect from the variability within each sample. Compare the relative variability in mean values with the variability within the sample groups with that seen in Figure 3-2.

Is this variability in $\bar{x}_j$ consistent with the variability observed within the individual sample x_i values?

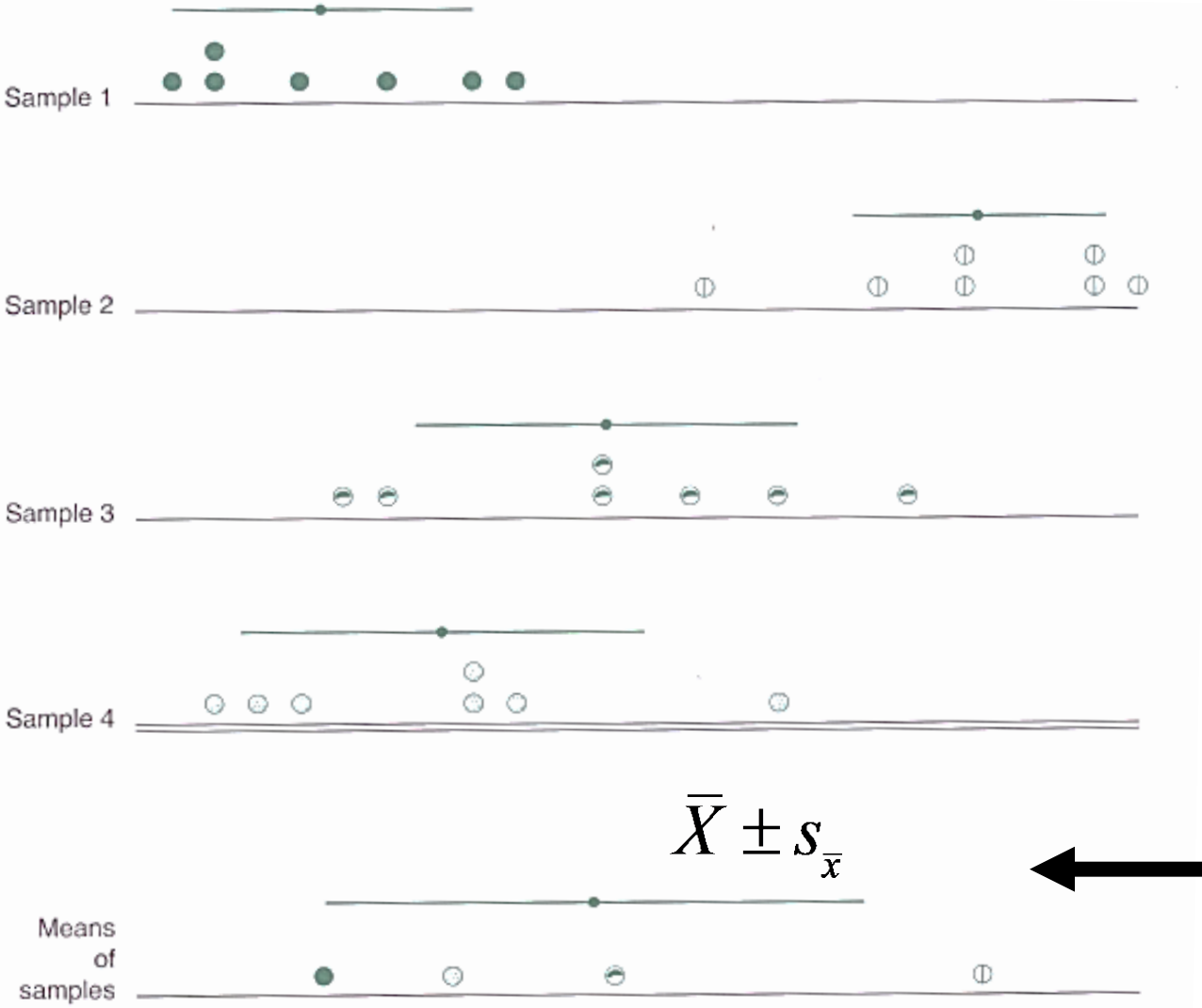
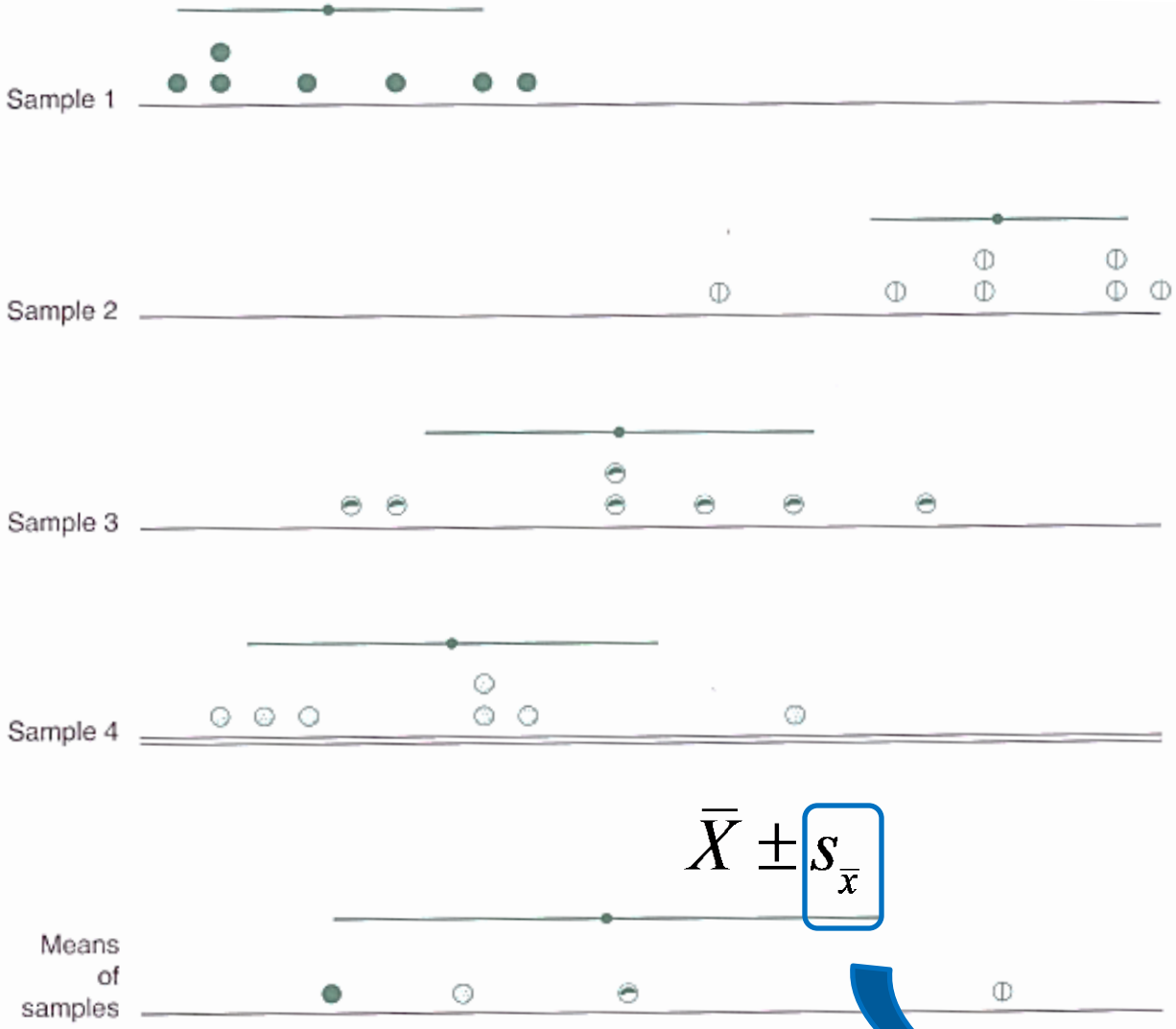


FIGURE 3-4. When the mean of even one of the samples (sample 2) differs substantially from the other samples, the variability computed from within the means is substantially larger than one would expect from examining the variability within the groups.

Is this variability in $\bar{x}_j$ consistent with the variability observed within the individual sample x_i values?



$$X \sim N(\mu, \sigma^2)$$

$$\bar{X} \pm s_{\bar{x}}$$

$$\bar{X} \sim N\left(\mu, \left(\frac{\sigma}{\sqrt{n}}\right)^2\right)$$

FIGURE 3-4. When the mean of even one of the samples (sample 2) diff substantially from the other samples, the variability computed from within the means is substa lly larger than one would expect from examining the variability within the groups.

II. ANOVA

A. Two Estimates of Population Variance

Assume $CO \equiv X \sim N(\mu, \sigma^2)$: single population

Null Hypothesis H_0: No difference in CO among C, P, S, and N groups

Estimate $\boldsymbol{\sigma^2}$ in two ways; if all subjects are drawn from same population, the two estimates will be (roughly) the same

1. Within-groups estimate of variance

$$\bar{x}_C = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad s_C^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \bar{x}_C\right)^2 \qquad n = 7$$

All groups have same number of subjects, $n = 7$

Also calculate $\bar{x}_P, \bar{x}_S, \bar{x}_N$ & s_P^2, s_S^2, s_N^2

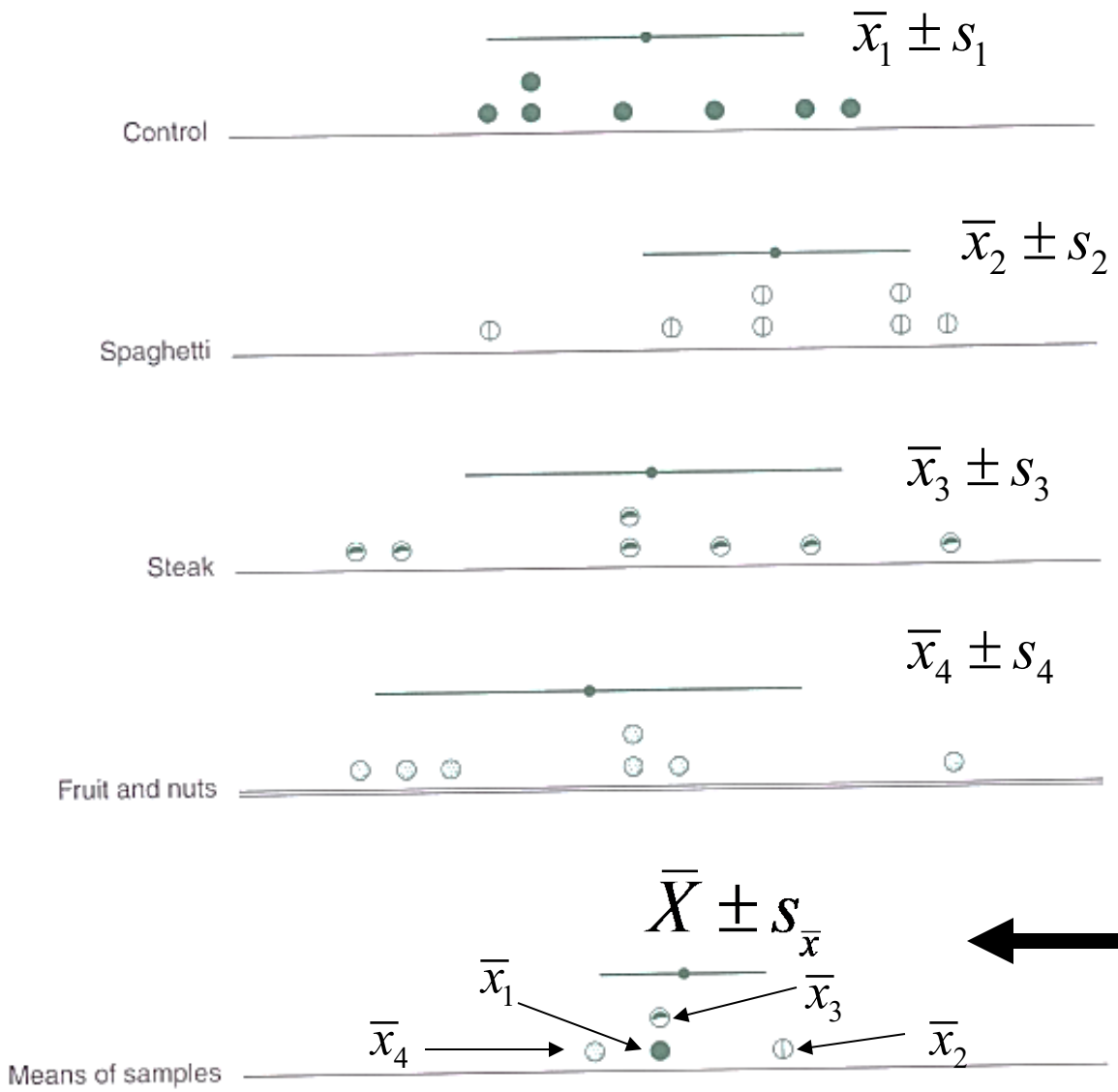
If all samples are from the same population (as per H_0), then

s_C^2 s_P^2 s_S^2 s_N^2 are all estimates of σ^2

Get a better estimate of σ^2 $\qquad m = 4 =$ number of groups

$$s_{wit}^2 = \frac{1}{m}\left(s_C^2 + s_P^2 + s_S^2 + s_N^2\right)$$

Estimate of population variance using within-group variances



Question:

Are these differences in CO due to diet or are these due to random variation in CO in the population?

Is this variability in $\bar{x}_j$ consistent with the variability observed within the individual sample x_i values?

2. Between-groups estimate of variance

First from: $\bar{x}_C,\ \bar{x}_P, \bar{x}_S, \bar{x}_N$ calculate std. dev. of sample means

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m}\bar{x}_i \qquad s^2_{\bar{x}} = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2 \qquad m = 4$$

Then recall: $\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}} \qquad \sigma^2 = n\sigma^2_{\bar{x}}$

All groups must have same number of subjects, *n*, to use this formulation

Now from samples: $s^2_{bet} = ns^2_{\bar{x}}$

Estimate of population variance calculated from variance between group means

B. The F Statistic

$$F = \frac{s^2_{bet}}{s^2_{wit}}$$

F is a random variable defined by a p.d.f. $f(F)$

What is a "Big" value of F???

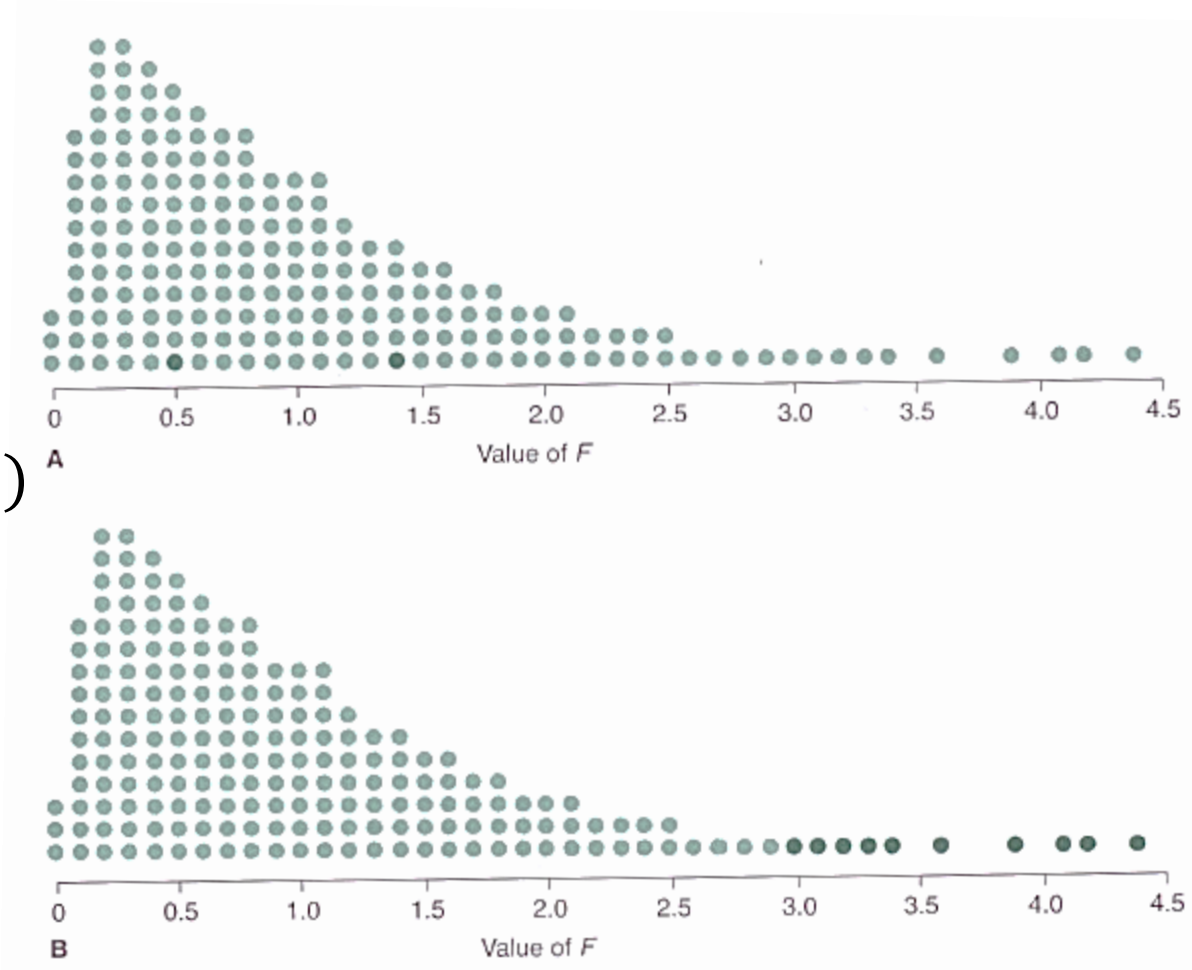


FIGURE 3-6. (A) Values of F computed from 200 experiments involving four samples, each of size 7, drawn from the population in Figure 3-1. **(B)** We expect F to exceed 3.0 only 5% of the time when all samples were, in fact, drawn from a single population. *(continued)*

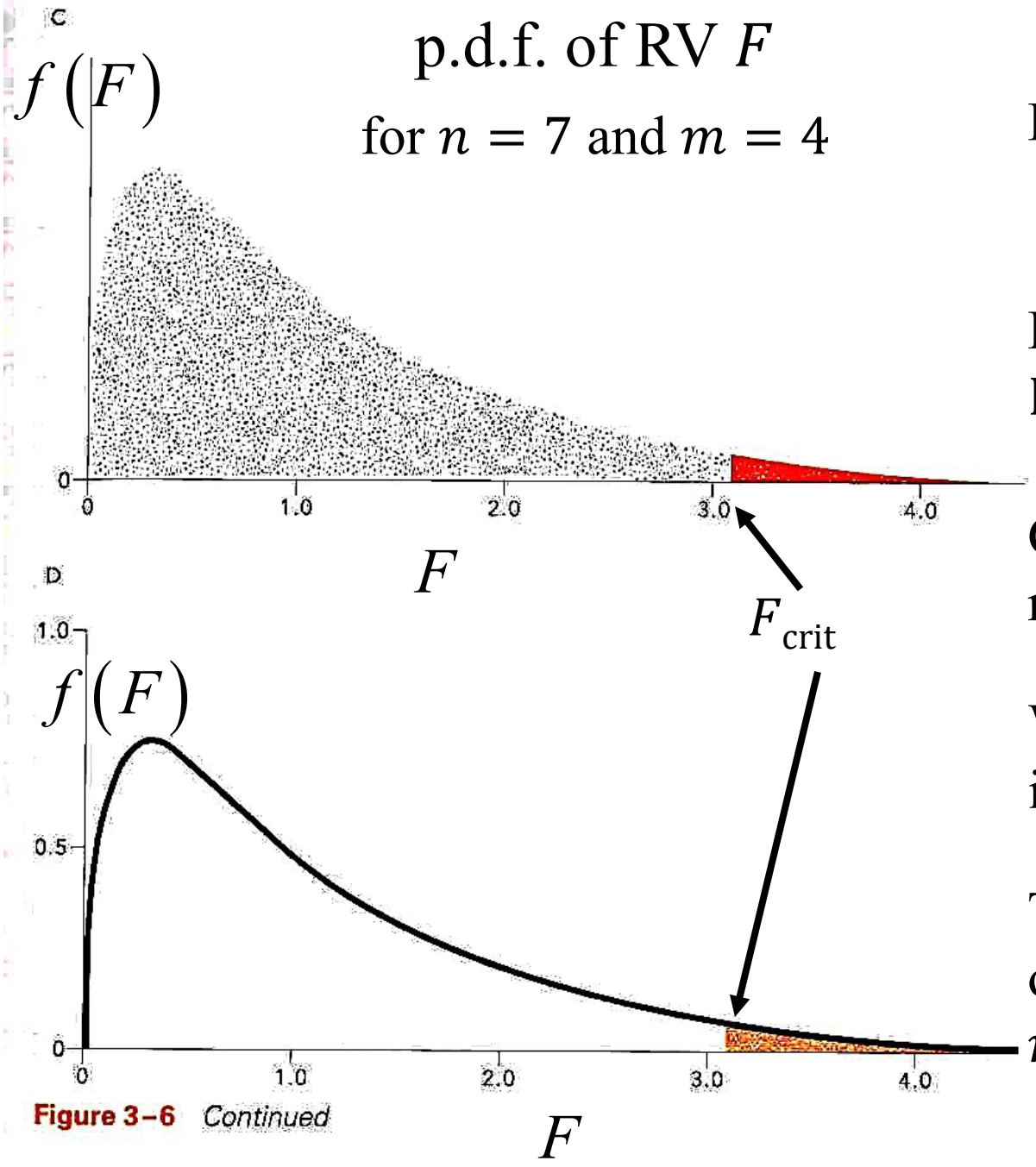


Figure 3–6 *Continued*

p.d.f. of RV F

for $n = 7$ and $m = 4$

H_0: No difference in CO among groups.

Define F_{critical} as follows:
$P(F > F_{\text{crit}}) = 0.05$

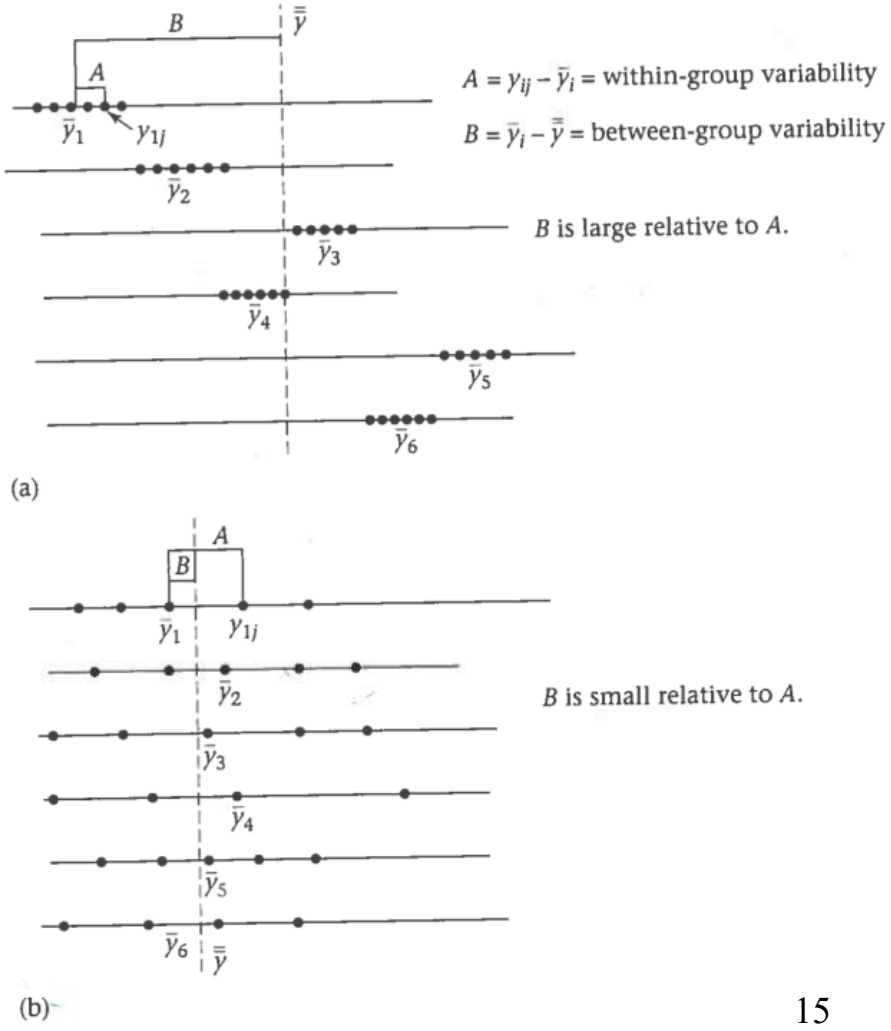
Conclusion: if $F > F_{\text{crit}}$ reject H_0 with $P < 0.05$

Where P is probability of incorrectly rejecting H_0

The value of F_{crit} depends on n (# subjects) and m (# groups)

Comparing Variance Within and Between Groups

- Want to determine whether or not means of different groups are same (i.e., if groups are drawn from same population or not)
- Two sources of variance (each can be used to estimate the population variance):
  - Variation within groups: A
  - Variation between groups: B
- When B is large relative to A, conclude that one or more group means is/are not equal to the others (individuals not all drawn from same population)
- When B is small relative to A, conclude all individuals are drawn from same population (all group means are same)



III. Examples: A. Cell Phones & Sperm Motility

TABLE 3-2. Sperm Motility (%)

Observed Cell Phone Use	Number of Subjects (n)	Mean	Standard Deviation
Control (no cell phone use)	40	68	6
Low use (<2 h/d)	40	65	8
Medium use (2 to 4 h/d)	40	54	11
High use (>4 h/d)	40	45	16

(Glantz, *Primer of Biostats*)

H_0: No difference in sperm motility among cell-phone usage groups.

n: number of subjects per group $= 40$
m: number of groups $= 4$
Mean & standard deviation unit: %
Variance unit: $\%^2$

1. Within-groups estimate of variance

$$s^2_{wit} = \frac{1}{m}\left(s^2_{control} + s^2_{low} + s^2_{medium} + s^2_{high}\right) = \frac{1}{4}\left(6^2 + 8^2 + 11^2 + 16^2\right) = 119.3\%^2$$

2. Between-groups estimate of variance

$$\bar{X} = \frac{1}{m}\left(\bar{x}_{control} + \bar{x}_{low} + \bar{x}_{medium} + \bar{x}_{high}\right) = \frac{1}{4}\left(68 + 65 + 54 + 45\right) = 58\%$$

$$s_{\bar{x}} = \sqrt{\frac{\left(\bar{x}_{control} - \bar{X}\right)^2 + \left(\bar{x}_{low} - \bar{X}\right)^2 + \left(\bar{x}_{medium} - \bar{X}\right)^2 + \left(\bar{x}_{high} - \bar{X}\right)^2}{m-1}} = \sqrt{\frac{\left(68-58\right)^2 + \left(65-58\right)^2 + \left(54-58\right)^2 + \left(45-58\right)^2}{4-1}} = 10.44\%$$

$$s^2_{bet} = ns^2_{\bar{x}} = 40(10.44\%)^2 = 4360\%^2$$

3. The F Statistic

$$F = \frac{s_{bet}^2}{s_{wit}^2} = \frac{4360}{119.3} = 36.53$$

degrees of freedom

$$\nu_n = m - 1 = 4 - 1 = 3$$

$$\nu_d = m(n-1) = 4(40-1) = 156$$

$$F_{\text{crit}} = F_{\alpha}(\nu_n, \nu_d) = F_{0.05}(3,156)$$

$$F_{0.05}(3,120) = 2.68 > F_{\text{crit, }0.05} > F_{0.05}(3,\infty) = 2.60$$

$$F_{0.01}(3,120) = 3.95 > F_{\text{crit},0.01} > F_{0.01}(3,\infty) = 3.78$$

$$F = 36.53 > 3.95 > F_{crit}$$

Conclusion: Reject H_0 with $P < 0.01$

There is a difference in sperm motility among the cell-phone usage groups

Exercise

Burnout is a term that loosely describes a condition of fatigue, frustration, and anger manifested as a lack of enthusiasm for and feeling of entrapment in one's job. This situation can arise when treating people who have serious diseases. In recent years, AIDS has joined the list of diseases that may have a negative impact on professionals serving people suffering from this disease. To investigate whether there were differences in burnout associated with caring for people who have AIDS compared with other people who have serious diseases, J. López-Castillo and coworkers administered the Maslach Burnout Inventory questionnaire to health professionals working in four clinical units: infectious disease, hemophilia, oncology, and internal medicine in Spain (see table). (Ninety percent of the people in the infectious disease and 60% of the people in the hemophilia unit were HIV positive.) Are there differences in burnout scores among the health professionals working in these different units?

TABLE 3-8. The Maslach Burnout Inventory Questionnaire

	Infectious Disease	Hemophilia	Oncology	Internal Medicine
Mean	46.1	35.0	44.4	47.9
Standard deviation	16.1	11.1	15.6	18.2
Sample size (*n*)	25	25	25	25

- Assumptions of ANOVA
- Extension to case of unequal sample sizes
- Two-Way ANOVA
- Repeated Measures ANOVA